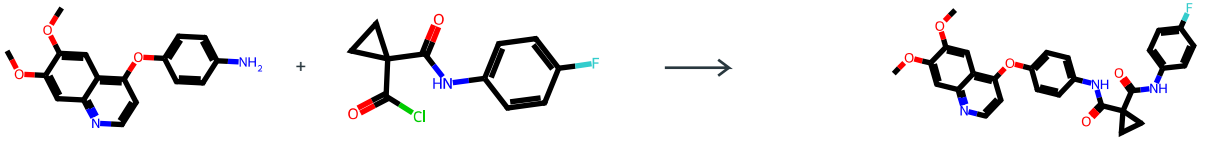
Performed experimental precedent for generated complete reactions

Selected cases; complete net chemistry, not experimentally established electron-arrow sequences.

1. Tier A · US20160022662A1:cabozantinib-free-base:acid-chloride-coupling

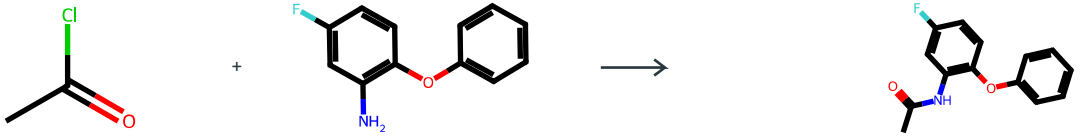


K2CO3; THF/water; addition not exceeding30C; reported completion approximately20min; precipitation at20-25C approximately10h

2. Tier A · WO2018104954A1:example1:acid-to-acid-chloride

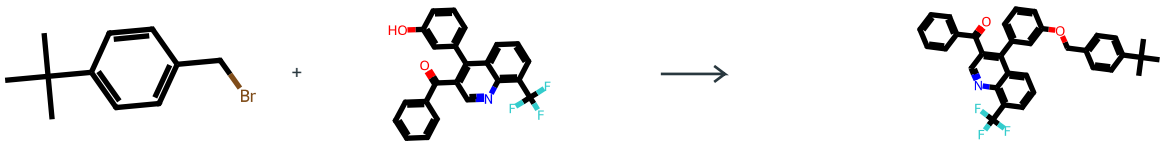
Example1: acid112.9g, thionyl chloride180.4g, THF600mL,25-30C,5h; then used in situ for coupling. No acid-chloride isolated yield or independent characterization is given.

3. Tier A · US8188116B2:example11:phenoxyaniline-acetylation



Pyridine 15 mL; aniline 5.14 g/25 mmol; acetyl chloride 2.3 mL/33 mmol added at 0 C, then reflux 1 h. Chromatography gives 5.6 g, 90%; product NMR and GC-MS are reported.

4. Tier A · US7576215B2:example489:quinoline-phenol-benzylation



Example 489 explicitly refers to Example 478: cesium carbonate (source notation CsCO3) in DMF, 60 C for 14 h, then reversed-phase HPLC. The 33% yield belongs to Example 478's nitrobenzyl analogue and is NOT assigned to Example 489.

Source/training overlap unknown; an experimentally documented transformation is not a tested whole route.